TCS NQT PYQ QUESTIONS

Company: TCS NQT
Round: Online
Year: 2025
Difficulty: High
Topic: Mixtures and Alligation

Problem Description:

A vessel contains 40 litres of a mixture of milk and water in the ratio 3:1. 8 litres of the mixture is drawn off and replaced with water. This process is repeated once more (drawing off 8 litres of the new mixture and replacing it with water).

Find the final ratio of milk to water in the vessel.

Options:

A) 3:2
B) 4:5
C) 12:13
D) 5:6

Correct Answer:

C) 12:13

Approach to Solve This Problem:

STEP 1:

Initial mixture: 40 litres, milk:water = 3:1, so milk = 30 litres and water = 10 litres.

STEP 2:

For repeated replacement, the standard result is: milk left after n operations = initial milk × $(1 – x/V)^n$, where x is the volume removed each time and V is the total volume.

Here x = 8, V = 40, n = 2, so the fraction remaining each time is (1 – 8/40) = 0.8.

STEP 3:

Milk after 2 operations = 30 × $(0.8)^2$ = 30 × 0.64 = 19.2 litres.

Water = 40 – 19.2 = 20.8 litres.

STEP 4:

Ratio of milk to water = 19.2 : 20.8 = 192 : 208 = 12 : 13.

CORRECT ANS = 12:13